# The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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## 1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification ($Fx$), while the invariant control forests encode properties atemporally ($xF$).

## 2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[ \boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

## 3 Distributed Multi-Directory Agent Initializations

Multi-agent domain isolation requires true file-system sandboxing to prevent VRAM memory leakages. Each independent neuro-symbolic agent $a_k \in \mathcal{A}$ maintains its unique background logic constraints inside a dedicated disk directory path:

$$a_k \equiv \langle \mathcal{V}_k, \mathcal{K}_k, \mathcal{F}_k \rangle \tag{2}$$

## 4 Theory Movement and Graph-of-Trees Ontology Revision

We formalize the transport and reconciliation of local ontologies across distinct agents via the Model Context Protocol (MCP) using a Graph-of-Trees composition model.

### 4.1 Step 4: MCP Tool Payload Encapsulation

Once Dr. Rasiowa's consensus logic satisfies the implication boundary ($\mathcal{M} \models \text{COMBINE_AND_EXECUTE}$), the source agent $a_1$ tokenizes and serializes its decision forest memory $\mathcal{F}_1$ into a standardized binary payload $\mathcal{P}_{1 \to 2}$ via the MCP tool interface. The binary object encodes the nested conditional rules derived from the source agent's empirical exemplars.

### 4.2 Step 5: Forest Grafting and Structural Integration

Upon receiving the payload, the target agent $a_2$ performs a non-destructive ontology revision. Let $\mathcal{T}_2$ represent the decision tree structure of the target agent, and $\mathcal{E}_{\text{exit}}$ represent the terminal exit branch node mapping to un-handled features (the `False` path of local kinematics). The grafting operator $\oplus$ isolates the root pointer of the incoming source tree $\mathcal{T}_1$ and binds it directly to the exit branch, establishing an expanded, unified conditional domain:

$$\mathcal{T}_2^{\text{unified}} = \mathcal{T}_2 \oplus_{\mathcal{E}_{\text{exit}}} \mathcal{T}_1 \tag{3}$$

This composition ensures that if an incoming query vector evaluates to `False` over local kinematics features, the feature evaluation token slides seamlessly into the migrated Searle social world vertices. This allows cross-domain execution passes to unroll under zero-uncertainty parameters, preserving Mesarovic's Attainable Space $V'$.